

RETAIL SALES VISUALIZATION, RELATIONSHIP ANALYSIS & BUSINESS INSIGHTS USING SUPERSTORE SALES DATASET

OBJECTIVE

The Superstore Sales Dataset was analyzed using Python and Seaborn to understand sales performance, profitability, discount impact, and regional variations. Bar plots, box plots, scatter plots, and a correlation heatmap were used to identify important business patterns.

ANALYSIS AND KEY FINDINGS

Analysis	Key Finding	Business Insight
Sales by Category	Sales contribution varies across categories.	High-sales categories can be prioritized for inventory and marketing decisions.
Profit by Category & Sub-Category	Profitability differs across products; some sub-categories may generate low or negative profit.	Loss-making products should be reviewed for pricing and discount policies.
Sales & Profit Distribution	Box plots show variation and extreme values in sales and profit.	Outliers should be investigated to identify unusual transactions or business opportunities.
Discount vs Profit	Profit changes with different discount levels, with higher discounts potentially reducing profitability.	Excessive discounting should be controlled to protect profit margins.
Regional Performance	Profit contribution varies between regions.	Regional pricing, product mix, and discount strategies should be analyzed separately.
Correlation Analysis	Sales, Quantity, Discount, and Profit show different levels of relationships.	Correlation helps identify variables that are useful for further business analysis.

DISCOUNT AND PROFIT ANALYSIS

The scatter plot of Discount versus Profit was used to understand whether discounting affects profitability. The analysis indicates that highly discounted transactions can have lower or negative profits. Average profit at different discount levels can be used to identify a suitable discount threshold. However, discount is not the only factor affecting profit; product cost, selling price, and quantity may also influence the result.

MAJOR BUSINESS INSIGHTS

- High sales do not always mean high profit.
- Some sub-categories can create profit leaks despite generating sales.
- Excessive discounts can negatively affect profitability.
- Regional performance varies, requiring region-specific strategies.
- Extreme sales and profit values should be investigated before removing them.
- The relationship between Sales, Quantity, Discount, and Profit provides useful information for business decision-making.

RECOMMENDATIONS

- **Pricing:** Review products with high sales but low profit.
- **Discounts:** Establish suitable discount limits for low-margin products.
- **Products:** Investigate consistently loss-making sub-categories.
- **Regions:** Identify successful regional strategies and improve weaker regions.
- **Monitoring:** Regularly track sales, discount, and profit together through dashboards.

CONCLUSION

The analysis demonstrates that visualization provides deeper business insights than numerical summaries alone. The Superstore dataset shows the importance of balancing sales growth with profitability. A controlled discount strategy, improved product pricing, and region-wise performance monitoring can help reduce profit losses and support sustainable business growth.